

Graded lab 3

a. Calculate the Sum of Squares of First n Natural Numbers Using Stack and BP

Write an 8086 Assembly Language program that performs the following tasks:

1. Accepts an integer value n.
2. Passes n to a procedure named SUMSQUARES using the stack.
3. The SUMSQUARES procedure computes the sum of squares of the first n natural numbers.
 - o Formula: $S = 1^2 + 2^2 + 3^2 + \dots + n^2$
4. The result is returned in the AX register.
5. The procedure must use the BP register to access its parameter from the stack.

b. Write an 8086 Assembly Language procedure named ISPRIME that checks if a given integer is a prime number.

- n is the input integer.

Requirements:

1. The parameter n must be passed to the procedure via the stack.
2. The procedure must determine whether n is a prime number.
 - o A prime number is only divisible by 1 and itself.
 - o If n is prime, set the AL register to 1; if not, set it to 0.
3. The procedure should not use the BP or SP registers for managing local data or parameters.